속도는벡터 · 소개영상 — Veo 3.1 26 segment 정본 (slide 14장 × script 3:28)

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작성 2026-05-27 23:58 KST · 5/28 12:00 LearnUs 마감 약 12 시간 남음

★ **자료 base**: `submission/제출완료/속도는벡터_기말발표.pptx` (14 슬라이드 v3 PDF carry) + `submission/제출완료/기말발표_스크립트.pdf` (5 페이지 한국어 narration)

★ **사용자 명시 5/27 KST**: Full 3:30 · 26 segment · 스크립트 그대로 carry · slide 14장 narrative 보존. 영상 안 텍스트 = **영어 only** · audio narration = **한국어** · 한국어 자막 = **합성 단계 SRT 후처리** (별도).

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0. 영상 사양

항목	값
총 길이	**3:28 (208 초)**
Segment	**26 개 × 8 초** (Veo 3.1 segment)
해상도	1920 × 1080 (16:9 가로, 1080p)
프레임	30 fps
오디오	Veo 3.1 native audio — 한국어 narration + 효과음 + 옅은 ambient
영상 안 텍스트	**영어 + 숫자 only** (한국어 X)
자막	**합성 단계 SRT 후처리** (DaVinci Resolve / Flow / Clipchamp)
비주얼 톤	**흰 배경 + navy `#1E3A5F` (주요) + cyan `#3DAFC1` (강조) + coral `#D14F4F` (대조)** · clean academic · slide v3 톤 carry
모델	Veo 3.1 - Quality (Flow `에이전트 설정`)

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1. 26 segment timeline (slide 14장 × 1-3 seg)

Slide	단계	seg	시간	핵심 메시지
1	표지	1 (1A)	0:00-0:08	인덱스 부재 시 Adaptive Sampling 개선 · 팀명
2	배경	3 (2A·2B·2C)	0:08-0:32	VAQ 정의 + SQL 예시 + 4 테이블 조인
3	배경 ★ video	3 (3A·3B·3C)	0:32-0:56	카디널리티·선택도 + 10,000× 응답 시간 차이 (ref 영상 톤)
4	배경	2 (4A·4B)	0:56-1:12	pgvector 33.3% · VBASE 50% · DuckDB-vss 100%
5	방법	2 (5A·5B)	1:12-1:28	Adaptive Sampling 4 단계 + 본 연구 = 1 STEP 표본 추출
6	방법	1 (6A)	1:28-1:36	Research Question + baseline ↔ 샘플링 탐색
7	방법	2 (7A·7B)	1:36-1:52	baseline · 단독 대체 · 결합 (CORE)
8	방법	2 (8A·8B)	1:52-2:08	1,508가지 조합 (5 데이터셋 × 4 변인 × 16 method)
9	방법	1 (9A)	2:08-2:16	7 paradigm 13 method
10	결과 ★	2 (10A·10B)	2:16-2:32	**89.1%** Q-error 우위 (1,344/1,508)
11	적용 ★	3 (11A·11B·11C)	2:32-2:56	5,677/977.6/983.5 ms + plan **58.3% → 94.9%**
12	적용 ★ video	2 (12A·12B)	2:56-3:12	**5.70× 응답 시간 단축** (최종 엔진 결합 방식)
13	적용	1 (13A)	3:12-3:20	Future Work (GROUP A · GROUP B)
14	마무리	1 (14A)	3:20-3:28	감사합니다 + Q&A

총 26 segment × 8s = 208s = 3:28

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2. Veo 3.1 prompt — 26 segment (Flow 직접 복붙)

각 prompt = 영문 visual + 한국어 audio narration. Flow chat 에 그대로 복붙 + 화살표 + Approve.

Slide 1 — 표지

Segment 1A · 0:00-0:08 — 도입·팀

```
8-second clean academic title-card animation, 1920x1080, white background.

VISUAL: A clean white background with a thin horizontal line at the center-top. Below the line, two-row large bold

CAMERA: Static framing. Title text reveals with a smooth scale-up.

LIGHTING: Soft even white background.

MOOD: Calm, academic opening, the talk begins.

KOREAN AUDIO NARRATION (calm academic male voice): "안녕하세요, 속도는 벡터 팀, 발표 시작하겠습니다. 인덱스가 없는

SOUND EFFECTS: Soft chime when title reveals.

MUSIC: Very subtle ambient pad, low volume.

IMPORTANT: All visible text in the video must be in English or numbers only. No Korean text visible.
```

Slide 2 — VAQ 정의 (3 segment)

Segment 2A · 0:08-0:16 — VAQ 정의 (분석가 질문)

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top center: bold title in navy "Vector-Augmented Query (VAQ)". Below the title,

CAMERA: Static framing. Title fades in first, then analyst icon, then question bubble, then the 4 table icons.

LIGHTING: Soft even white background.

MOOD: Setting the scene, the analyst poses a question.

KOREAN AUDIO NARRATION (calm academic male voice): "벡터 증강 분석 쿼리, VAQ. 관계형 조건과 벡터 유사도가 한 SQL 

SOUND EFFECTS: Subtle UI chimes for each element appearing.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 2B · 0:16-0:24 — SQL 코드 + 벡터 술어

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. In the center, a code block fades in with monospace terminal-style typography. T
"SELECT l_orderkey, o_orderdate, o_shippriority"
"FROM   customer, orders, lineitem, partsupp_deep"
"WHERE  c_mktsegment = 'HOUSEHOLD'"
"AND    c_custkey = o_custkey"
"AND    l_orderkey = o_orderkey"
"AND    ps_partkey = l_partkey"
"AND    ps_embedding distance to query under 0.86"
The last two lines (vector predicate) are highlighted in cyan color, and a small cyan label points to it: "Vector

CAMERA: Static framing. Code lines reveal sequentially with a typing effect. The cyan highlight pulses gently.

LIGHTING: Soft even white background.

MOOD: Technical reveal, the query structure exposed.

KOREAN AUDIO NARRATION (calm academic male voice): "예시 쿼리에서 관계형 조건과 벡터 유사도 임계값 영점 팔육이 한

SOUND EFFECTS: Soft typing sound as code appears.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 2C · 0:24-0:32 — 4 테이블 조인 + optimizer plan

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Left side: a stylized icon of a database optimizer (a small gear icon labeled "O

CAMERA: Static framing. Arrows and decisions appear sequentially.

LIGHTING: Soft even white background.

MOOD: Mechanism reveal, the optimizer's decisions.

KOREAN AUDIO NARRATION (calm academic male voice): "옵티마이저는 벡터 조건의 결과 행 수, 카디널리티를 추정해 전체

SOUND EFFECTS: Subtle clicks as decisions appear.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 3 — 카디널리티·1만× (3 segment)

Segment 3A · 0:32-0:40 — 카디널리티·선택도 정의

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: bold title "Cardinality and Selectivity". Two definition boxes side by side
Left box: large bold "CARDINALITY" + below it "= number of rows that satisfy a condition". Small example: "1,000,0
Right box: large bold "SELECTIVITY" + below it "= ratio of matched rows". Small example: "10,000 / 1,000,000 = 1 p

CAMERA: Static framing. Two boxes reveal simultaneously.

LIGHTING: Soft even white background.

MOOD: Definition, foundational concept.

KOREAN AUDIO NARRATION (calm academic male voice): "카디널리티는 조건을 만족하는 결과 행 수, 선택도는 그 비율입니다

SOUND EFFECTS: Soft chimes for each definition.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 3B · 0:40-0:48 — 추정 오차 → plan 변화 (ref 영상 톤)

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Side-by-side two panels with a thin vertical divider. Left panel (coral-red): to

CAMERA: Static framing. Both panels reveal simultaneously.

LIGHTING: Soft even white background.

MOOD: Contrast, mechanism reveal.

KOREAN AUDIO NARRATION (calm academic male voice): "카디널리티 추정 한 곳이 틀리면 실행 계획이 통째로 바뀝니다. 길

SOUND EFFECTS: Two soft chimes, one per panel.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 3C · 0:48-0:56 — 10,000× 응답 시간 차이 hero

```
8-second clean academic explainer animation, 1920x1080, white background, dramatic number reveal.

VISUAL: A clean white background. In the center, a massive bold number reveals with a smooth scale-up animation: "

CAMERA: Static framing. Number reveals with smooth scale-up, bars animate.

LIGHTING: Soft even white background.

MOOD: Dramatic problem reveal, the cost.

KOREAN AUDIO NARRATION (calm academic male voice): "카디널리티 한 곳이 잘못되면 같은 쿼리가 최대 일만 배까지 느려

SOUND EFFECTS: Soft bass impact when "10,000x" appears, then a long whooshing sound.

MUSIC: Very subtle ambient pad with slight emphasis at the number reveal.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 4 — 기존 시스템 한계 (2 segment)

Segment 4A · 0:56-1:04 — pgvector·VBASE·DuckDB 33.3·50·100%

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Existing Systems - Fixed Ratio Estimate". Below, three vertical pane
Panel 1 (light gray box): system name "pgvector" + massive bold "33.3%" + small text below "always assumes 1/3 of
Panel 2 (light gray box): system name "VBASE" + massive bold "50%" + small text below "always assumes 1/2 of rows"
Panel 3 (light gray box): system name "DuckDB-vss" + massive bold "100%" + small text below "essentially full scan

CAMERA: Static framing. Three panels reveal sequentially.

LIGHTING: Soft even white background.

MOOD: Problem statement, all systems get it wrong.

KOREAN AUDIO NARRATION (calm academic male voice): "기존 시스템은 데이터 분포와 무관하게 고정 비율로 카디널리티를

SOUND EFFECTS: Three quick "tick" chimes as each panel appears.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 4B · 1:04-1:12 — Exqutor 도입 + index 없음 분기

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Exqutor - Two Paths". A horizontal flow diagram: a central node labe
Top branch: arrow to a small box labeled "Vector Index Present -> Use Index Directly".
Bottom branch: arrow to a larger highlighted box (cyan border) labeled "Vector Index Absent -> Adaptive Sampling".
Below the diagram, a centered emphasized text in dark navy: "This research focuses on no-index path".

CAMERA: Static framing. Diagram reveals top to bottom, then the focus highlight pulses on the bottom branch.

LIGHTING: Soft even white background.

MOOD: Pivot to research scope, the no-index case.

KOREAN AUDIO NARRATION (calm academic male voice): "Exqutor 는 인덱스가 있으면 인덱스를, 없으면 적응적 표본 추출을

SOUND EFFECTS: Subtle whoosh as the diagram reveals.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 5 — Adaptive Sampling 4 단계 (2 segment)

Segment 5A · 1:12-1:20 — 4 단계 흐름

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Adaptive Sampling - 4 Steps". Below, a horizontal flow diagram of 4

CAMERA: Static framing. Boxes reveal sequentially left to right with the Our Focus badge appearing last.

LIGHTING: Soft even white background.

MOOD: Method reveal, the algorithmic flow.

KOREAN AUDIO NARRATION (calm academic male voice): "적응적 표본 추출은 표본 추출, 카디널리티 추정, 큐 에러 측정,

SOUND EFFECTS: Soft ticks as each box appears.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 5B · 1:20-1:28 — N 갱신 과정 + 본 연구 개입

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: small label "Sample Size N adjusts over time". Below, a horizontal sequence

CAMERA: Static framing. Boxes reveal left to right, then the focus contrast appears.

LIGHTING: Soft even white background.

MOOD: Pivoting from N to selection, the intervention.

KOREAN AUDIO NARRATION (calm academic male voice): "기존 적응적 표본 추출은 N 을 얼마로 둘지에 집중했습니다. 본

SOUND EFFECTS: Soft chimes for N adjustments, emphasis chime for the focus contrast.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 6 — Research Question (1 segment)

Segment 6A · 1:28-1:36 — Research Question + baseline ↔ 분포 인지

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: small label "RESEARCH QUESTION" in purple monospace. Below, a large bold se
Left box: title "BASELINE", below a small scatter pattern of cyan dots distributed uniformly (random Bernoulli), c
Right box: title "DISTRIBUTION-AWARE", below a small pattern showing 4 groups of organized dots (clusters/curves/g

CAMERA: Static framing. Question reveals first, then both boxes simultaneously.

LIGHTING: Soft even white background.

MOOD: Research question framing, contrast of approaches.

KOREAN AUDIO NARRATION (calm academic male voice): "본 연구의 질문. 카디널리티 추정을 더 잘할 수 있는 표본 추출 방

SOUND EFFECTS: Soft chime for the question, two chimes for the boxes.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 7 — 세 추정 방식 (2 segment)

Segment 7A · 1:36-1:44 — baseline + 단독 대체 (X)

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Three Estimation Modes Compared". Left side: panel "BASELINE" with s

CAMERA: Static framing. Left panel reveals first, then middle panel with the X strike appearing dramatically.

LIGHTING: Soft even white background.

MOOD: Negative control reveal, replacement alone is unstable.

KOREAN AUDIO NARRATION (calm academic male voice): "베이스라인 단독, 새 method 단독 대체. method 단독은 특정 쿼리

SOUND EFFECTS: Soft chime for baseline, a sharp click for the X strike.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 7B · 1:44-1:52 — 결합 (CORE) + 산술 평균

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Center: large box with a cyan border labeled "COMBINED (CORE)" with subtitle "(b

CAMERA: Static framing. Two arrows merge into one at the center.

LIGHTING: Soft even white background.

MOOD: The core method reveal, combination.

KOREAN AUDIO NARRATION (calm academic male voice): "본 연구는 베이스라인 추정값과 분포 인지 method 추정값을 산술 

SOUND EFFECTS: Two soft whooshes merging, a confirmation chime when FINAL appears.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 8 — 1,508 조합 (2 segment)

Segment 8A · 1:52-2:00 — 5 데이터셋 × 4 변인

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Two adjacent panels:
Left panel (purple border): title "DATASETS (5)" + 5 small rows each with a small icon and label: "DEEP - image se
Right panel (purple border): title "VARIABLES (4)" + 4 small rows: "SCALE FACTOR: sf = 1, 10, 100", "SELECTIVITY: 

CAMERA: Static framing. Both panels reveal simultaneously, then rows fade in sequentially.

LIGHTING: Soft even white background.

MOOD: Experimental scope, the rigor.

KOREAN AUDIO NARRATION (calm academic male voice): "실험은 다섯 개 데이터셋과 네 조작변인 위에서 진행되었습니다. 

SOUND EFFECTS: Subtle ticks as rows appear.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 8B · 2:00-2:08 — 1,508 조합 hero

```
8-second clean academic explainer animation, 1920x1080, white background, large number reveal.

VISUAL: A clean white background. Center top: small calculation strip "5 datasets x sf {1, 10, 100} x sel {0.001,

CAMERA: Static framing. Equation reveals with smooth scale-up. Grid fades in last.

LIGHTING: Soft even white background.

MOOD: Scale reveal, the measurement plane.

KOREAN AUDIO NARRATION (calm academic male voice): "총 천 오백 팔 가지 조합에서 베이스라인과 결합 방식을 직접 비교

SOUND EFFECTS: A confirmation chime when "1,508" appears, then soft ticks for the grid.

MUSIC: Very subtle ambient pad with slight emphasis at the number.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 9 — 7 paradigm 13 method (1 segment)

Segment 9A · 2:08-2:16 — 7 paradigm 분류

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "13 Methods Grouped Into 7 Paradigms". Below, a grid of 7 small panel
Top row (4 panels): "P1 Space-Filling Curve - Hilbert, Z-order", "P2 Dimension Reduction - PCA, rSVD", "P3 Classic
Bottom row (3 panels): "P5 Clustering - KMeans, GMM", "P6 Streaming - chao_weighted", "P7 Hashing - md5 prefix".
On the left side, a smaller box labeled "BASELINE - uniform Bernoulli" with a small dot pattern.

CAMERA: Static framing. Panels reveal sequentially.

LIGHTING: Soft even white background.

MOOD: Method taxonomy, the candidate space.

KOREAN AUDIO NARRATION (calm academic male voice): "분포 인지 표본 추출 13 method 를 일곱 패러다임으로 분류했습니다

SOUND EFFECTS: 7 quick soft ticks.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 10 — 89.1% Q-error 우위 (2 segment)

Segment 10A · 2:16-2:24 — hero "89.1%" reveal

```
8-second clean academic explainer animation, 1920x1080, white background, hero number reveal.

VISUAL: A clean white background with a deep navy rectangular hero block centered. Inside the navy block, a massiv

CAMERA: Static framing. The hero block fades in, then the number reveals with a smooth scale-up.

LIGHTING: Soft even white background.

MOOD: Dramatic headline result reveal.

KOREAN AUDIO NARRATION (calm academic male voice): "결과입니다. 전체 천 오백 팔 측정 중 천 삼백 사십 사 셀에서 결

SOUND EFFECTS: A soft bass impact when the navy block appears, a confirmation chime for "89.1%".

MUSIC: Very subtle ambient pad with slight emphasis at the number.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 10B · 2:24-2:32 — baseline 1.4582 vs 결합 1.4019 막대

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top center: small label "Average Q-error - lower is better". Below, two horizont
Top bar (gray): "Baseline" label on the left, bar extends to "1.4582" on the right.
Bottom bar (cyan, highlighted): "Combined" label on the left, bar slightly shorter, ending at "1.4019" on the righ
A thin vertical guide line marks "1.30" on the left edge and "1.50" on the right edge of the bars (axis range).

CAMERA: Static framing. Both bars fill in animation.

LIGHTING: Soft even white background.

MOOD: Quantitative validation, the precision.

KOREAN AUDIO NARRATION (calm academic male voice): "평균 큐 에러는 베이스라인 일 점 사 오 팔 이, 결합 방식 일 점 

SOUND EFFECTS: Soft fill sound for each bar.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 11 — 엔진 latency + plan 선택 (3 segment)

Segment 11A · 2:32-2:40 — 3 막대 latency

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Engine Latency - DEEP sf=10, 156 plans". Below, three horizontal bar
Bar 1 (gray, long): label "pgvector default" + bar extending to "5,677 ms (1.0x)" on the right.
Bar 2 (navy, short): label "Baseline (original)" + bar much shorter ending at "977.6 ms (5.77x faster)".
Bar 3 (cyan, short): label "Combined (our research)" + bar similar length to baseline, ending at "983.5 ms (5.70x
Below the bars, small caption in dark navy: "Baseline and Combined - essentially equal latency".

CAMERA: Static framing. Three bars fill sequentially.

LIGHTING: Soft even white background.

MOOD: Engine validation, speedup is real.

KOREAN AUDIO NARRATION (calm academic male voice): "피지벡터 기본은 오천 육백 칠십 칠 밀리초. 베이스라인은 구백 칠

SOUND EFFECTS: Three fill sounds, one per bar.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 11B · 2:40-2:48 — plan 선택 91 → 148 (58.3% → 94.9%)

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Optimal Plan Selection". Two large circles side by side:
Left circle (dark navy): large bold number "91 / 156" centered + label below "Baseline: 58.3%".
Right circle (cyan, brighter): large bold number "148 / 156" centered + label below "Combined: 94.9%".
Between them, a small orange box with bold "+57 plans" inside, with subtitle "91 -> 148".
Below the circles, small caption: "Optimal plan = fastest actual execution time".

CAMERA: Static framing. Left circle fills first, then right circle, then the +57 box pops in.

LIGHTING: Soft even white background.

MOOD: Structural advantage, plan stability.

KOREAN AUDIO NARRATION (calm academic male voice): "최적 plan 선택률에서 베이스라인은 156 중 91, 결합 방식은 156

SOUND EFFECTS: Two fill sounds for circles, a confirmation chime for "+57 plans".

MUSIC: Very subtle ambient pad with slight emphasis.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 11C · 2:48-2:56 — 최적 plan 정의 + 결론

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Latency Same, Plan Stability Improved". Below, a horizontal flow dia

CAMERA: Static framing. Diagram flows left to right, then the conclusion text reveals.

LIGHTING: Soft even white background.

MOOD: Conclusion of engine validation, balanced result.

KOREAN AUDIO NARRATION (calm academic male voice): "결합 방식은 평균 응답 시간을 베이스라인보다 나쁘게 만들지 않으

SOUND EFFECTS: Three soft tones for the flow boxes, a confirmation chime for the conclusion.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 12 — 최종 엔진 작동·5.70× 단축 (2 segment)

Segment 12A · 2:56-3:04 — 최종 엔진 작동 흐름

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Final Engine - Combined Method Applied". Below, a horizontal flow wi

CAMERA: Static framing. Boxes reveal sequentially with arrows.

LIGHTING: Soft even white background.

MOOD: Engine application, the full pipeline.

KOREAN AUDIO NARRATION (calm academic male voice): "최종 엔진에서 결합 방식 카디널리티를 적용하면 옵티마이저가 정

SOUND EFFECTS: Soft ticks as each box appears.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Segment 12B · 3:04-3:12 — 5.70× 단축 hero

```
8-second clean academic explainer animation, 1920x1080, white background, hero number reveal.

VISUAL: A clean white background with a deep navy rectangular hero block centered. Inside the navy block, a massiv

CAMERA: Static framing. Hero block fades in, then number reveals.

LIGHTING: Soft even white background.

MOOD: Final speedup result, the engineering win.

KOREAN AUDIO NARRATION (calm academic male voice): "결합 방식 최종 엔진은 피지벡터 기본 대비 오 점 칠 영 배 응답 

SOUND EFFECTS: Soft bass impact when navy block appears, confirmation chime for "5.70x".

MUSIC: Very subtle ambient pad with slight emphasis.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 13 — Future Work (1 segment)

Segment 13A · 3:12-3:20 — GROUP A + GROUP B

```
8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top: title "Future Work - Two Directions". Two adjacent panels side by side:
Left panel (orange border): label "GROUP A - Scope Expansion" + below 4 small tag chips arranged in 2x2: "scale up
Right panel (cyan border): label "GROUP B - History-Aware Adaptive Sampling" + below a small flow diagram: "Past q

CAMERA: Static framing. Both panels reveal simultaneously.

LIGHTING: Soft even white background.

MOOD: Looking forward, the next steps.

KOREAN AUDIO NARRATION (calm academic male voice): "향후 연구는 두 갈래입니다. 검증 범위 확장과 과거 쿼리 피드백을

SOUND EFFECTS: Two soft chimes for each panel.

MUSIC: Very subtle ambient pad.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

Slide 14 — 마무리 (1 segment)

Segment 14A · 3:20-3:28 — 감사합니다 + Q&A

```
8-second clean academic closing animation, 1920x1080, white background.

VISUAL: A clean white background with a thin horizontal line at the center-top. Below the line, two-row large bold

CAMERA: Static framing. Closing text reveals with smooth scale-up. Footer fades in.

LIGHTING: Soft even white background.

MOOD: Warm closing, gratitude.

KOREAN AUDIO NARRATION (calm academic male voice, warm tone): "이상으로 발표를 마치겠습니다. 감사합니다."

SOUND EFFECTS: A warm closing chime.

MUSIC: Very subtle ambient pad resolves to silence at the end.

IMPORTANT: All visible text in the video must be in English or numbers only.
```

3. 합성·자막 후처리 workflow

3.1 Flow 진행

1. https://labs.google/fx/ko/tools/flow → 기존 프로젝트 진입 (또는 새 프로젝트)
2. `에이전트 설정` → 동영상 모델 **Veo 3.1 - Quality** + 16:9 + 1x 확정
3. 우측 chat 입력창에 위 §2 의 Segment 1A → 14A prompt 26 개 순차 복붙
4. 각 prompt 당 약 60-120s generation (Veo 3.1 Quality)
5. 각 segment 출력 mp4 확인 + 다운로드 (파일명 `segment_<NN>.mp4`)

3.2 합성 (3 옵션)

옵션 A: Flow 자체 Storyboard (추천) — 좌측 "장면" tab → 26 segment 순서 배치 → fade through white 0.2s transition → Export 1080p mp4

옵션 B: DaVinci Resolve (무료, 자막 burn-in 포함) — 26 mp4 import → timeline 배치 → Cross Dissolve 0.2s → **자막 layer (SRT import)** → Deliver YouTube 1080p

옵션 C: Clipchamp (Windows 무료) — 26 mp4 import → Fade transition → Export 1080p

3.3 한국어 자막 SRT 후처리

영상 안 자막 없음 (Veo 한글 깨짐 회피). 합성 단계에서 SRT burn-in:

SRT 파일 생성 (한국어 narration text + segment timing):

```
1
00:00:00,000 --> 00:00:08,000
안녕하세요, 속도는 벡터 팀, 발표 시작하겠습니다.
인덱스가 없는 환경에서 적응적 표본 추출을 개선하는 연구입니다.

2
00:00:08,000 --> 00:00:16,000
벡터 증강 분석 쿼리, VAQ. 관계형 조건과 벡터 유사도가
한 SQL 안에 함께 들어가는 쿼리입니다.

(... 26 segment 동일 형식 ...)
```

전체 SRT = 본 md 의 각 segment KOREAN AUDIO NARRATION 텍스트 + segment 시작·종료 (8초 단위).

자막 스타일: Apple SD Gothic Neo · 흰 텍스트 + navy `#1E3A5F` 그림자 · 화면 하단 중앙 · 폰트 크기 36-40pt.

3.4 YouTube + QR + 포스터

1. YouTube Unlisted 업로드 (제목: "속도는벡터 — 캡스톤 종합설계 결과 (연세대 BDAI 연구실)")
2. URL → qr-code-generator.com PNG (500×500)
3. 포스터 우측 footer QR placeholder 갱신 → PDF re-export
4. LearnUs 5/28 12:00 마감 전 final 제출

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4. 핵심 정량 수치 (Veo prompt 안 직접 사용)

위치	수치	의미
Segment 3C	10,000x	Slide 3 hero (Wrong vs Correct estimate latency 차이)
Segment 4A	33.3% · 50% · 100%	Slide 4 fixed-ratio (pgvector · VBASE · DuckDB-vss)
Segment 5B	N=385	Slide 5 initial sample size
Segment 8B	1,508	Slide 8 measurement cells
Segment 10A	89.1%	Slide 10 hero (1,344/1,508 better)
Segment 10B	1.4582 vs 1.4019	Slide 10 average Q-error
Segment 11A	5,677 / 977.6 / 983.5 ms	Slide 11 engine latency
Segment 11B	91/156 = 58.3% → 148/156 = 94.9% · +57 plans	Slide 11 plan selection
Segment 12B	5.70x	Slide 12 hero (combined vs pgvector default speedup)

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5. 환각 회피 룰 (carry · 변경 X)

- 영상 안 텍스트 = 영문 + 숫자 only · 한국어 절대 X (Veo 한글 자모 깨짐)
- audio narration = 한국어 (Veo TTS) · 발표 스크립트 carry
- 한국어 자막 = 합성 단계 SRT 후처리 burn-in (DaVinci / Flow / Clipchamp)
- 비주얼 톤 = 흰 배경 + navy `#1E3A5F` 주요 + cyan `#3DAFC1` 강조 + coral `#D14F4F` 대조
- 슬라이드 14장 narrative carry (Slide 1 → 14 순서)
- 정량 수치 carry (89.1% · 5.70× · 94.9% · 58.3% · 5,677 / 977.6 / 983.5 ms · 1,508 · 10,000×)

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6. 자료 carry

자료	경로
★ 발표 스크립트 (5 페이지 한국어)	`submission/제출완료/기말발표_스크립트.pdf`
★ 발표 슬라이드 (14장 pptx)	`submission/제출완료/속도는벡터_기말발표.pptx`
슬라이드 PDF v3 (14 페이지 carry)	`submission/_drafts/archive/2026_05_26_postsubmit/deck/속도는벡터_기말발표_v3.pdf`
슬라이드 PNG 14장 (시각 reference)	`/tmp/slides/slide-01.png ~ slide-14.png` (본 세션 생성)
ref 영상 (32초, 무음)	`submission/_drafts/archive/2026_05_26_postsubmit/video/무제.mp4`

작성 2026-05-27 23:58 KST · 5/28 12:00 LearnUs 영상·포스터 마감 약 12 시간 남음 · 26 segment Veo 3.1 explainer animation 3:28 · slide 14장 + script base · 영문 visual + 한국어 audio narration + SRT 합성 후처리.